

Zsh: The Ultimate Alternative to Bash



A shell is the interface to the operating system and the services provided by it. The default shell on any operating system is the bash shell. However, there are more and better alternatives to the bash shell. Zsh is one of them.

The terms *terminal*, *tty*, *console* and *shell* are used synonymously. There is nothing wrong with that but these words mean a lot of different things. They represent pieces of hardware that were used to enter data into the computer in earlier times, as the machines back then used to be very big and data entry was done using terminals, which were also known as teletypewriters (hence, *tty*). Physically, they were known as consoles (from the furniture point of view). The terminal (console) in our computers these days is an application which is similar to the terminal of past days but it doesn't run on a big machine; instead, it runs on the top of the kernel in our machine. Coming to shell, it is a command interpreter that makes the use of the terminal very easy by interpreting certain characters in a specified manner—for example, tab completion in *bash*, which was absent in the older shells. In the same way, *bash* lacks some features which are covered by *zsh*.

Figure 1 shows that the shell works as an interpreter.

Different types of shell

Sh: The Bourne shell, called *sh*, is one of the original shells, developed for UNIX computers by Stephen Bourne at AT&T's Bell Labs in 1977. Its long history of use means many software developers are familiar with it. It offers features such as input and output redirection, shell scripting with string and integer variables, and condition testing and looping.

Bash: The popularity of *sh* motivated programmers to develop a shell that was compatible with it, but with several enhancements. Linux systems still offer the *sh* shell, but *bash* (the Bourne-again shell), which is based on *sh*, has become the new default standard. One attractive feature of *bash* is its ability to run *sh* shell scripts, unchanged. Shell scripts are complex sets of commands that automate programming and maintenance chores; being able to reuse these scripts saves programmers time. Conveniences not available with the original Bourne shell include command completion and a command history.